

LUC3D: 3D annotation and proofreading GUI for pose estimation and multi-animal tracking

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Abstract

Animal behavior unfolds in three dimensions over time, yet accessible scientific tools for analyzing it in this native form remain scarce. Innovation in 2D pose estimation methods have increased the ability of neuroscientists to track behaviors over time but can often lead to ambiguous estimates. However, uncertainty from 2D pose methods can be refined with multi-camera approaches to allow for more accurate 3D estimates. This paper presents a browser-based 3D GUI for labeling, proofreading, and multi-animal tracking that requires no installation. In addition, we present multiple comprehensive datasets of proofread multi-camera 3D triangulated pose data of multi-animal mouse social behavior. Also, a protocol for generating 3D consistent labels is presented and a novel cross-view ReID tracking algorithm for proofreading multi-animal multi-camera pose data is made available.

Introduction

Advances in deep learning have made 2D pose estimation using convolutional neural networks an accessible and popular method for quantifying behavior ([Pereira et al., 2020](#), [Pereira et al., 2022](#)). However, 2D pose estimation has limitations with respect to occlusion, distance, projective ambiguity and depth. The solution to these myriad of problems caused by the limitations of 2D is to use multiple camera views to enable 3D pose estimation methods ([Marshall et al., 2022](#), [Karashchuk et al., 2021](#)). Here we present LUC3D, a browser-based GUI that brings together SLEAP's 2D tracking pipeline and 3D multi-camera functionality for multi-animal multi-camera recordings. The LUC3D GUI allows for the user to leverage 3D triangulation and reprojection to streamline the labeling process for multiple camera views with careful consideration of user experience. In addition to new software tools, this paper presents two comprehensive datasets: the BMimica-11M two-mouse data repository and the SLAP-2M data repository. Many popular 2D datasets lack environmental enrichment, by adding levels of enrichment this dataset highlights the need for 3D pose estimation. This makes the SLAP-2M dataset an ideal candidate for benchmarking new models for behavioral segmentation (both supervised and unsupervised) and 3D pose estimation.

Results

A browser-based tool resolves one identity per animal in every camera

Multi-camera pose recording produces a per-camera problem and a cross-camera problem. The per-camera problem, finding the keypoints, is solved by existing detectors. The cross-camera problem, deciding which detection in one view is the same animal as which detection in another, has no standard tool, and until it is solved there is no 3D reconstruction and nothing to proofread.

LUC3D addresses that problem in a web browser with no installation and no build step (Figure 1A). The three stages it contributes are cross-view re-identification, triangulation, and proofreading of the resulting 3D; the per-view detector is not ours, and the application consumes SLEAP predictions.

Figure 1B shows the operation on a single frame of an eight-camera recording of three mice. The detector supplies 24 detections across the eight views, carrying 20 distinct track names, and those names are per camera and arbitrary: the same animal is track 89 in one view and track 83 in another. After re-identification the frame holds three identities, one per animal in every view, with every detection assigned. Figure 1C triangulates that frame and fills all 45 of its 3D keypoints, and shows the reconstruction placed at one camera's own pose and field of view so that it can be compared directly against that camera's video.

Figure 1D places the tool among seven existing packages. Reprojection-aided labelling is not new, and Label3D and JARVIS both provide it. What no other tool in the table combines is annotation-time cross-view identity, multi-animal support, a 3D proofreading viewport, and running in a browser without installation.

Two labelled views are enough, because the reprojection into the rest is usually already correct

Manual multi-camera annotation scales with the number of cameras: a labeller places every keypoint in every view. If the 3D point can be solved from two views, the remaining views can be filled by reprojection, and the labeller only has to correct those reprojections that land badly. The saving is therefore governed by one measurable quantity, the fraction of reprojections that a labeller would reject.

We measured that fraction on 50 proofread sessions, 1,277,424 keypoints (Figure 2C). A 3D point solved from two views and projected into a view that was not labelled lands a median 4.32 px from that view's own detection, with 94.6 per cent of reprojections within 10 px and 99.68 per cent within 20 px. At a 10 px acceptance tolerance only 5.4 per cent of reprojections need touching at all.

Figure 2B turns that into placements per animal per frame. On the five-camera rig we measured, accepting reprojections at 10 px reduces manual placements from 75 to 32, a 2.3-fold saving, and the saving grows with rig size because the labelled two views stay fixed while the reprojected remainder does not. Only one rig size was measured, so every other point on that panel is the model rather than a measurement, and the panel draws it that way.

The cost of solving from only two views is geometric, and Figure 2D shows it directly. The error of a two-anchor solve is set by the angle the anchor pair subtends at the animal and follows the depth-uncertainty relation k over $\sin \theta$ with k equal to 1.52 mm, so the practical advice is not to label two views but to label two views that are far apart. A pair subtending 15 degrees gives roughly 12 mm of error where a pair at 30 degrees gives under 3 mm, against a floor of 1.2 mm when all five views contribute.

A greedy per-camera assignment matches exhaustive enumeration at a millionth of the cost

The published approach to annotation-time cross-view association enumerates every possible grouping of detections into identities, triangulates and reprojects each, and keeps the grouping with the lowest reprojection error. That is A factorial per view and therefore A factorial raised to the power C per frame, which is 32 hypotheses for two animals in five cameras and 1.9 hundred million for four animals in six (Figure 3C). LUC3D instead solves one assignment problem per camera and commits it before moving to the next, which costs C times A cubed (Figure 3A).

The question is whether the cheap procedure gives up accuracy, and Figure 3D answers it against proofread ground truth rather than against the other method. On all 137,671 frames where exhaustive enumeration could be run at all, the greedy grouping differs from ground truth on one frame and the exhaustive optimum on four. On the five frames where the two methods choose different groupings, ground truth agrees with the greedy choice on four of them. Optimising the reprojection objective harder does not buy accuracy here, because the objective itself is what runs out: the lowest-reprojection-error grouping is not always the correct grouping.

Figure 3F prices the two procedures on identical detections. The gap is a factor of about a million in the four-animal, six-camera configuration, which exhaustive enumeration could not run at all. That point is drawn as a bound rather than a measurement, and the bound is generous to the published method twice over, since it removes a relabelling symmetry that method does not exploit and prices the remainder at the cheapest rate measured anywhere in the sweep. Even so, one second of 50 fps video would cost more than a day.

Two limits belong with this result. Exhaustive enumeration is undefined unless every camera holds exactly as many detections as there are animals, so it was run only on the 69 per cent of frames that satisfy that, and the excluded frames are the occluded ones, which is to say exactly the frames association finds hardest. And 89 per cent of the eligible frames are the easiest configuration, two animals in five cameras. The

equivalence therefore holds where the published method can run, and is not a claim about the frames where it cannot.

Figure 3E asks what the three-dimensional term in the association cost contributes, by sweeping it against the two-dimensional term over 24 combinations on eight complete sessions. The two weights turn out to matter only through their ratio: every pair of combinations sharing a ratio returns identical scores. The term is necessary. With it switched off entirely, identity switches rise to 40,984 and cross-view IDF1 falls to 0.595. Bringing it up to parity with the two-dimensional term removes 98 per cent of those switches, leaving 696 and lifting IDF1 to 0.750. Beyond parity the two metrics behave differently: switches continue to fall slowly, reaching 272 at the largest weights tested, while IDF1 varies between 0.71 and 0.76 without a clear trend. The shipped default sits in that region. It is a safe choice rather than one this sweep selected, and the figure marks it as the default and nothing more.

Triangulation accuracy is set by geometry, not by the solver

Having established the correspondence, the remaining question is what governs the accuracy of the resulting 3D point. Figure 4 separates three candidates: the number of contributing views, the treatment of a view that disagrees, and the choice of solver.

View count dominates. Scored in a camera the solve never saw, error falls from 4.32 px with two cameras to 3.34 px with four (Figure 4B). Rejecting a disagreeing view matters on the tail rather than in the bulk: when the worst-fitting view sat more than 10 px from the all-view solution, dropping it moves the 3D point by a median 7.2 mm, against 1.1 mm when that view was already within 3 px (Figure 4C). Those large corrections are 1.6 per cent of keypoints, which the panel prints under the axis, so the 7.2 mm figure describes a small and identifiable minority rather than typical behaviour.

The solver, by contrast, barely matters. Across four solvers, two from this work and two from aniposelib, the entire spread on a held-out camera is 3.11 to 3.34 px (Figure 4D). Neither library's non-linear refinement earns its cost out of sample, and aniposelib's linear solve is the lowest of the four. We report that plainly: our solvers are 4.4 and 2.8 times faster than the corresponding aniposelib paths (Figure 4E), and that is a speed result, not an accuracy result.

Two measurement decisions carry this figure and are worth stating. Scoring in a camera excluded from the solve is essential, because our refinement minimises reprojection error in the cameras it was given, so any metric computed on those cameras is one the method optimises. And distance to the proofread reconstruction cannot be used to rank solvers at all: the refinement moves the estimate in a direction essentially uncorrelated with the direction to the reference, and adding a displacement orthogonal to an existing error always increases the distance, so that metric would report that the refinement had moved regardless of whether it improved.

Cross-view identity makes a social behaviour measurable that no single camera can see

The preceding figures are about the tool. Figure 5 asks what the tool is for, by measuring a behaviour that only exists once the 3D is available.

Rearing is defined by height above the floor, and a pair interaction by the distance between two animals. Neither quantity exists in a single camera view. In the example display of Figure 5A every camera sits between 58 and 76 degrees above the animals, and the two noses are 93 to 106 px apart in every one of the five views, so no view recovers the vertical and no view distinguishes the animals' separation from their projection.

Rearing in these pairs is coupled, and the coupling requires proximity (Figure 5G). Taking every rearing onset by one animal and reading out whether the other is rearing, the probability is 2.9 times chance at the onset itself and peaks at 4.1 times half a second later, which is about the time it takes a mouse to get up. When the animals are more than two body lengths apart the same measurement is flat at 1.05, and a circular-shift null that preserves each animal's rearing rate, bout structure and autocorrelation is flat at 0.99.

The second animal is responding to the first rather than coinciding with it, and a shared external drive is ruled out by the proximity split.

That coupling defines an event: both animals reared, within two body lengths, held for at least a quarter of a second. There are 539 such displays in 37 of the 56 sessions. They are brief and still. The animals hold the posture for a median 0.71 s while moving at 0.44 times their own baseline speed, with 94 per cent of displays below baseline (Figure 5E). A mutual upright posture at close range is the classic agonistic configuration, and the obvious description would be boxing, but animals moving at four tenths of their usual speed are not fighting. We therefore call it an upright display and go no further, since what the behaviour means cannot be settled by kinematics.

The display is not symmetric. One animal is up a median 0.37 s before the other joins (Figure 5C), which is 56 frames at this recording rate and one or two frames at 30 fps, so this measurement requires the high frame rate. Within a session, one animal starts most of the displays: pooled over all 539 displays, the session's leader starts 80 per cent of them, and that figure is stable at every session-inclusion threshold we tried (Figure 5F). Because the leader's share is by construction the larger of two shares, it cannot fall below one half, so the panel draws the distribution a fair coin would produce at each session size; 16 of the 24 sessions with at least five displays exceed it.

Three controls exclude the obvious alternatives. The leader is not simply the animal that rears more, since a session's initiation share is uncorrelated with its share of rearing time. It is not an artefact of the per-animal height threshold, since a single shared threshold and an absolute threshold in millimetres both return the same leader in almost every session. And it is not body size, since in the sessions where the initiator is the longer animal it still starts 79 per cent of displays. What the data support is an asymmetry that is stable within a session, and we make no claim about dominance, for which no assay was run.

Session difficulty removes keypoints rather than degrading them

Before comparing trackers it is worth knowing what varies across a corpus. Each of the 74 proofread SLAP-2M sessions carries a curator-assigned difficulty rating, and Figure 6C separates that rating's effect into two quantities that behave differently. From the easiest stratum to the hardest, the per-view miss rate rises 10.9-fold, from 5.3 to 57.9 per cent, while the mean error of the detections that do fire rises only 1.29-fold, from 3.67 to 4.72 px.

That asymmetry is favourable for multi-camera work, and it is the reason a multi-camera pipeline degrades gracefully. A keypoint missing in one view is recoverable from the others; a keypoint present but wrong in every view is not. Difficulty, as this corpus scores it, mostly produces the recoverable kind.

Figure 6D controls for the obvious confound, that difficulty tracks the number of animals, by re-running the same measurement within each animal count. The relationship survives, and the marginal miss rate by animal count alone is not even monotone, since it falls at four animals.

Only cross-view identity survives being pooled across cameras

Figure 7 compares LUC3D against per-camera trackers on identical, identity-stripped detections, so that what is compared is association rather than detection.

Within a single camera, all three methods are broadly competitive, and LUC3D's advantage over SLEAP is carried by the single-animal sessions where there is nothing to associate (Figure 7D). Pooled over all 74 sessions the difference in within-view IDF1 is 0.075 in our favour, but restricted to sessions with two or more animals it is 0.024 and does not clear a sign test, and in the two cells where cross-view association ought to help most, three and four animals, it is slightly negative. We state that rather than pooling it away.

The difference appears when identities are pooled across cameras (Figure 7A). Scoring all five cameras into one accumulator with a single global identity per animal, LUC3D's IDF1 is unchanged, 0.749 within view and 0.749 across, and drifts by at most 0.007 in any of 50 sessions. The per-camera trackers lose half to three quarters of theirs. This is close to arithmetic rather than a contest: a tracker that labels each

camera independently can have its labelling matched to the truth in at most one camera, so $1/C$ is a ceiling for such a method. The result is not that per-camera trackers are poor, but that per-camera identity is not the same object as cross-view identity, and that only the latter supports a 3D reconstruction or a proofreading pass that acts on all views at once.

The advantage does not depend on the animals being easy to tell apart. Under white bedding, where contrast against the coat is reduced, LUC3D's IDF1 falls by 0.012 while the per-camera trackers fall by 0.079 and 0.148, and the shared detector's own recall is essentially unchanged, which locates the loss in the association rather than in the detections (Figure 7B).

There is a cost, and Figure 7G reports it. LUC3D fragments tracks more than SLEAP does, by a mean of 24 fragmentations per camera-session, and SLEAP fragments less in 72 of the 74 sessions. A fragmentation is not an identity switch: the track breaks and is later resumed rather than being handed to the wrong animal. For a proofreading tool that is the preferable failure, since a break is visible and locally repairable while a switch propagates silently, but it is a real cost and it is the price of the same conservatism that produces the cross-view result.

Discussion

Online Methods

Datasets

Three multi-camera recordings were used. No animal procedures were performed for this work, and all three recordings are pre-existing datasets that had already been proofread in 3D.

The BMimica corpus consists of 56 sessions, each recording two mice with five synchronised cameras at 150 frames per second for approximately 20 minutes, giving 10,084,734 frames in total. A 15-node mouse skeleton was used throughout, comprising the nose, left and right ears, the tail-tip and tail base (denoted TTI), three intermediate tail nodes, the head, trunk and neck, and the left and right shoulders and haunches. Figures 2, 3E, 5 and 7A were measured on this corpus.

The SLAP-2M corpus consists of 84 sessions recording one to four animals with eight cameras at 50 frames per second, giving 1,954,440 frames, of which six cameras were proofread. The 74 sessions that carry a proofread 3D reconstruction were the ones measured, and every SLAP-2M measurement reported here uses the six proofread cameras. Each session also carries a curator-assigned difficulty rating and a black or white bedding label, which were used as strata in Figure 6 and as a control in Figure 7B. Figures 6 and 7B to 7G were measured on this corpus.

The third recording, HardFight, is a single eight-camera, three-mouse session at 60 frames per second, from which a 300-frame window was taken for Figure 1 and Figure 2A. It is the only recording that was built as a complete application session with its own calibration, which is why the panels driven through the application itself use it.

Camera calibration

Each rig ships a calibration file that gives, for every camera, its intrinsic matrix, its radial and tangential distortion coefficients and its extrinsic pose, in a TOML format compatible with aniposelib. The application reads these files unchanged, and no recalibration was performed for this work. Reprojection errors are reported throughout in each camera's own native, still-distorted pixels, because that is the space in which an annotator sees the error; a point is undistorted into ideal-pinhole coordinates only inside the linear solver, and the residual is measured after it is projected back. All 3D coordinates are expressed in the calibration's own metric frame, in millimetres. For Figure 5A the reconstructed pose was aligned to the calibration frame by a RANSAC-Procrustes fit, which retained 98.3 per cent of correspondences as inliers and left a residual of 1.32 mm.

Two-dimensional detection pool

Every benchmark reported here was run on a single shared detection pool from which all track and identity information had been removed, so that the comparisons isolate cross-view association and triangulation from the quality of the underlying detector. Each method under comparison received byte-identical input. Ground-truth identities were transferred onto that pool by matching detections to the proofread instances frame by frame and camera by camera at an intersection-over-union threshold of 0.5, so that a detector identity swap appears in the results as an error of the method under test rather than being silently absorbed. Detections were produced by SLEAP; the application consumes SLEAP .slp files and neither trains nor runs a detector of its own.

The measurements divide into two kinds, and they treat the recordings differently. The association and tracking results in Figures 3 and 7, and the behavioural analysis in Figure 5, were computed on complete sessions, every frame, with no subsampling. The per-keypoint geometric measurements in Figures 2, 4 and 6 were computed on a uniform sample of frames, taken every 200th frame for Figure 2, every 60th for Figures 4C to 4E, every 240th for Figure 4B and every 120th for Figure 6. The sample is uniform and was never selected on any property of the data, and it is large in absolute terms: 1,277,424 keypoints in Figure 2, 4,253,636 in Figures 4C to 4E, and 1,561,915 keypoint comparisons in Figure 6. Subsampling was necessary because these measurements are per-keypoint and combinatorial in the cameras. Figure 2 alone rests on 12,774,240 two-anchor solves and 38,322,720 held-out view measurements at its sampled density, and Figure 4B on 55,298,204 solves, so measuring at every frame would multiply those figures by two orders of magnitude. Two independent samplings of the same quantity agree closely: the 3D error by anchor count computed for Figure 4B at every 240th frame reproduces the same quantity computed for Figure 2 at every 200th frame to within 0.1 to 0.7 per cent at every camera count. The one place where the stride interacts with a method rather than merely with the sample size is temporal smoothing, and that is the reason the smoothing term in the comparison triangulator was disabled, as described below.

Cross-view association

For each frame the application maintains a set of 3D targets and solves one Hungarian assignment per camera, committing each camera's assignment to the target set before the next camera is solved, which costs $O(C \cdot A^3)$ for A animals and C cameras. The cost of pairing a target with a detection is summed over the skeleton nodes k with per-node weights w_k , where a weight of zero removes that node from the association entirely, and the sum is negated so that the solver minimises it. The cost has two terms. The first is a two-dimensional term on the distance between the detection and the target's reprojection into that camera, decayed by the age of the target. The second is a three-dimensional term on the perpendicular distance from the target to the ray back-projected from the detection, computed from the bare extrinsic matrix in normalised camera coordinates. Both thresholds are soft, in the sense that exceeding one drives its term negative rather than preventing the pairing outright. The shipped default weights are 1.0 on the two-dimensional term and 6.0 on the three-dimensional term.

The alternative method compared against in Figure 3 is our reimplement of the published per-frame procedure of Maree et al. (2024) rather than the authors' own code. It enumerates every grouping of detections into identities, of which there are $(A!)^C$ per frame, triangulates and reprojects each one, and keeps the grouping with the lowest reprojection error. That procedure is undefined on any frame where a camera does not hold exactly A detections, so a frame was treated as eligible only when every camera held exactly A , and ineligible frames were counted and reported rather than dropped. Of the 201,092 frames considered, 137,671 were eligible and 63,421 were skipped. The harness caps enumeration at one million hypotheses per frame, and the configuration of four animals in six cameras exceeds that cap by a factor of 191, so no frames of it were computed at all. Its cost in Figure 3F is therefore an arithmetic lower bound rather than a measurement: it takes the number of distinct hypotheses that remain after the $A!$ -fold global relabelling

symmetry is removed and prices them at the cheapest per-hypothesis rate measured anywhere in the sweep, and it is drawn with the open-marker convention this figure uses for a quantity that was not run.

Grouping quality in Figure 3D was scored by comparing partitions rather than labels. The grouping each method produced was compared with the ground-truth partition of the same matched detections, and a frame was counted as misgrouped when the two partitions differed, so that a consistent relabelling of the identities is not counted as an error.

The weight ablation in Figure 3E crossed a three-dimensional weight drawn from the set 0, 0.5, 1, 2, 4, 6, 8 and 12 with a two-dimensional weight drawn from 0.5, 1 and 2, giving 24 cells, and held every other tracker threshold at its default. Each cell was run on eight complete BMimica sessions, every frame, with identical detections. Cells that share a ratio of the two weights return identical IDF1 values and identical switch counts, and this collapse was verified rather than assumed, which is why the panel plots the ratio rather than the two weights separately.

Triangulation

The linear solver, which is the application's default, undistorts the observations into ideal-pinhole coordinates and minimises the algebraic error in closed form by direct linear transformation.

The non-linear solver minimises a geometric error in each camera's native, still-distorted pixels while holding the cameras fixed, using a soft-L1 loss followed by an L1 polish phase and initialised from the linear solution, with a backtracking guard that vetoes any step which raises the objective. Because the cameras are held fixed this is a triangulation and not a bundle adjustment, although the application's menu labels it "Bundle Adjustment". A true joint bundle adjustment, which would free the cameras as well as the structure, exists in the codebase and is deliberately not exposed in the interface, because rewriting a project's calibration would invalidate every 3D point already derived from it.

The application can optionally drop the view with the worst residual and re-solve the point from the remaining views. Figure 4C measures how far the estimate moves when it does so, stratified by how far the dropped view sat from the all-view solution.

The comparison triangulator in Figures 4D and 4E is aniposelib 0.7.2, the OpenCV and NumPy release that undistorts with `cv2.undistortPoints` and then solves each point with a singular value decomposition. This is the last release before the library was rewritten in JAX and the newest that anipose itself accepts; the build is asserted at run time and the measurement refuses to proceed against a JAX build, because the rewrite is a different program with different performance characteristics. The four solvers were compared in pairs matched by algorithm class, so that the linear closed-form solve of each library is compared with the other's and the non-linear cameras-fixed solve of each with the other's, and each comparison is therefore between two procedures doing the same amount of work. Enabling the library's RANSAC path costs 2,339 microseconds per keypoint, which is 83 times the default path, and it is reported here rather than drawn because a bar of that size would flatten every other bar in the panel. The temporal smoothing term in the non-linear path was disabled, because the input was sampled every 60th frame and consecutive entries are therefore more than a second apart, so the smoothing term would penalise real motion as though it were noise; with the smoothing left enabled the library's optimiser is worse than its own linear solve in all 50 sessions. Both variants were measured and deposited, and the panels draw the more favourable one.

Solve times exclude undistortion for every solver, because one library undistorts inside the triangulation call and the other outside it, and charging only one of them would measure where each library draws a function boundary rather than the cost of the solve; the excluded undistortion is 0.45 microseconds per keypoint for aniposelib and 1.17 for this work. Our solvers were timed in process under single-threaded Node 26 with `performance.now` around each call over 4,253,636 keypoints, and aniposelib with `time.perf_counter`, taking the best of three runs over 88,343 keypoints.

The accuracy measurements in Figures 2C and 4B are reference-free. A 3D point is solved from a subset of the cameras, projected into a camera that was not in that subset, and scored against that camera's

own raw detection, so no reference reconstruction enters the metric and neither solver optimises the quantity being reported. This estimator replaced an earlier one that measured distance to the proofread reconstruction, which cannot rank two solvers: the non-linear refinement moves the estimate in a direction essentially uncorrelated with the direction to the reference, with a median cosine of 0.004 at two cameras, and adding a displacement orthogonal to an existing error always increases the distance, so that measure reported that the refinement had moved whichever way it moved.

Behavioural analysis

A frame was counted as reared when the animal's neck height exceeded 0.75 of its own body length, where body length is that animal's median nose-to-tail-base distance across the session. Rearing bouts shorter than 0.25 s were discarded and gaps of 0.15 s or less within a bout were merged.

A mutual upright display was defined as both animals being reared with their tail bases within two body lengths of each other, held for at least 0.25 s. There are 539 such displays in 37 of the 56 BMimica sessions.

Because a display begins only when the second animal comes up, the onset of the display itself cannot identify which animal started it. The initiator was therefore identified from each animal's own rearing bout containing that onset, taking whichever animal's bout began earlier, and the lag reported in Figure 5C is the interval between the two bout onsets.

A session's leader, in Figure 5F, is whichever of the two animals initiated more of that session's displays. Because that statistic is the larger of the two shares it cannot fall below 0.5, so it was read against a simulated null rather than against a line at 0.5: for each session size, the 95th percentile of the larger share under a binomial distribution with probability one half was estimated from 20,000 draws. The figure reported in the text is pooled over displays rather than averaged over sessions, and the per-session statistics are restricted to sessions containing at least five displays.

Three controls were run on the initiator asymmetry. It is not explained by the rearing base rate, since a session's initiation share is uncorrelated with that animal's share of rearing time ($r = -0.01$). It is not an artefact of the per-animal height threshold, since re-running the detection with a single threshold for both animals returns the same leader in 22 of 24 sessions and an absolute 60 mm threshold returns the same leader in all 24. It is not body size, since in the eight sessions where the initiator is the longer animal it still starts 79 per cent of displays.

The coupling in Figure 5G was computed by taking every rearing onset by one animal and reading out the probability that the other animal was rearing at each lag within five seconds either side, divided by that other animal's own base rate. Onsets were split by the tail-base separation at the moment of onset, into those within two body lengths and those beyond. The null distribution was generated by circularly shifting the other animal's rearing time series, 24 shifts per ordered pair, which preserves that animal's rearing rate, its bout durations and its autocorrelation while destroying only the temporal alignment between the two animals; a reshuffle would additionally have destroyed the autocorrelation and would have made almost any structure appear significant. A session contributed a curve to a condition only if it supplied at least 20 onsets in that condition. The same measurement on the two-animal sessions of SLAP-2M gives 1.08 within two body lengths and 0.97 beyond, but that arena is 3.2 body lengths across against BMimica's 6.9, so the two conditions barely differ there and the claim is made for BMimica alone.

Tracking metrics

IDF1, identity switches and fragmentations were computed with the motmetrics library on the shared detection pool. Within-view IDF1 was computed separately for each camera and averaged. Cross-view IDF1 pools all cameras into a single accumulator with one global identity per animal; a method that labels each camera independently can be matched to the truth in at most one camera, so $1/C$ is a ceiling for such a method rather than a chance level, and chance is set by the number of animals rather than the number of cameras. A fragmentation was counted each time a tracked ground-truth track became untracked and was

later picked up again, which is a different event from an identity switch, in which the track continues but is assigned to the wrong animal. Error terms are reported as percentages of camera-frames. False negatives were measured but are not plotted, since they account for 98.8 to 99.3 per cent of every method's error budget and would draw three indistinguishable bars.

Statistics

The session is the unit of analysis throughout unless stated otherwise. Medians and interquartile ranges are used for the error distributions, which are right-skewed, and a band labelled as running from the 25th to the 75th percentile is an across-session spread of per-session medians rather than a confidence interval. Bootstrap confidence intervals at 95 per cent over sessions are used for the paired comparisons in Figures 7A, 7D and 7G. Sign tests are used for paired per-session comparisons of direction in Figure 7D. A Wilcoxon signed-rank test is used for the paired session-level comparison against a fixed value in Figure 5F. Binomial nulls were simulated rather than approximated wherever the statistic is a larger-of-two share, using 20,000 draws per session size. Circular-shift nulls were used for the temporal coupling in Figure 5G. No correction for multiple comparisons is applied across figures, and where a family of per-session tests is reported both the uncorrected and the Holm-corrected counts are given.

The application

LUC3D is a multi-view pose annotation and 3D proofreading interface that runs entirely in a web browser. It is served as static files with no installation and no build step, either from the hosted deployment or from any local static file server, and it keeps no state on a server, so a project is simply a set of local files that the user opens and saves.

It loads multi-camera video with frame-accurate WebCodecs decoding and scrubbing synchronised across every view, camera calibration in TOML or JSON, and SLEAP predictions, which are read lazily so that a prediction file larger than available memory can still be opened. Several recording sessions can be held open at once, each in its own dockable pane. A project can also be opened with no video at all, as a skeleton together with imported 3D points, in which case the whole duration of the reconstruction remains navigable.

Skeleton instances and their keypoints are created and edited directly on the video frames, and the node and edge definitions themselves can be edited inline. The reprojection-aided labelling protocol quantified in Figure 2 is available in the interface: the user labels two views, the 3D point is solved from those two alone, and the reprojection appears in every remaining view to be accepted or nudged.

A cross-view tracker groups the detections that belong to the same individual across views and assigns one identity per animal per frame, which is the capability measured in Figures 3 and 7. Its behaviour is exposed through a guided dialogue that offers per-node weights, per-view inclusion and the cost thresholds described above, so that the association can be tuned to a particular rig rather than accepted as a fixed policy. Tracks and identities are held per session and propagate to one another, so that a correction to either is reflected in the other.

Triangulation is offered as the linear solution and as the non-linear refinement described above, with optional rejection of an outlying view. The reconstruction is drawn in an interactive Three.js viewport that also renders the calibrated camera frustums, can be placed at any camera's own pose and field of view so that it can be compared directly against that camera's video, and can be exported as an MP4 video.

Reprojection error between the 3D triangulated points and the 2D labels is calculated for each keypoint and each view as an overlay on the frame the user is working on, which is the signal a proofreader works from; detections are drawn as solid marks and reprojections from the 3D as dotted ones. A timeline widget in the style of SLEAP carries frames, tracks and labelled regions for navigation, and the keyboard shortcuts are listed and can be rebound in the settings dialogue.

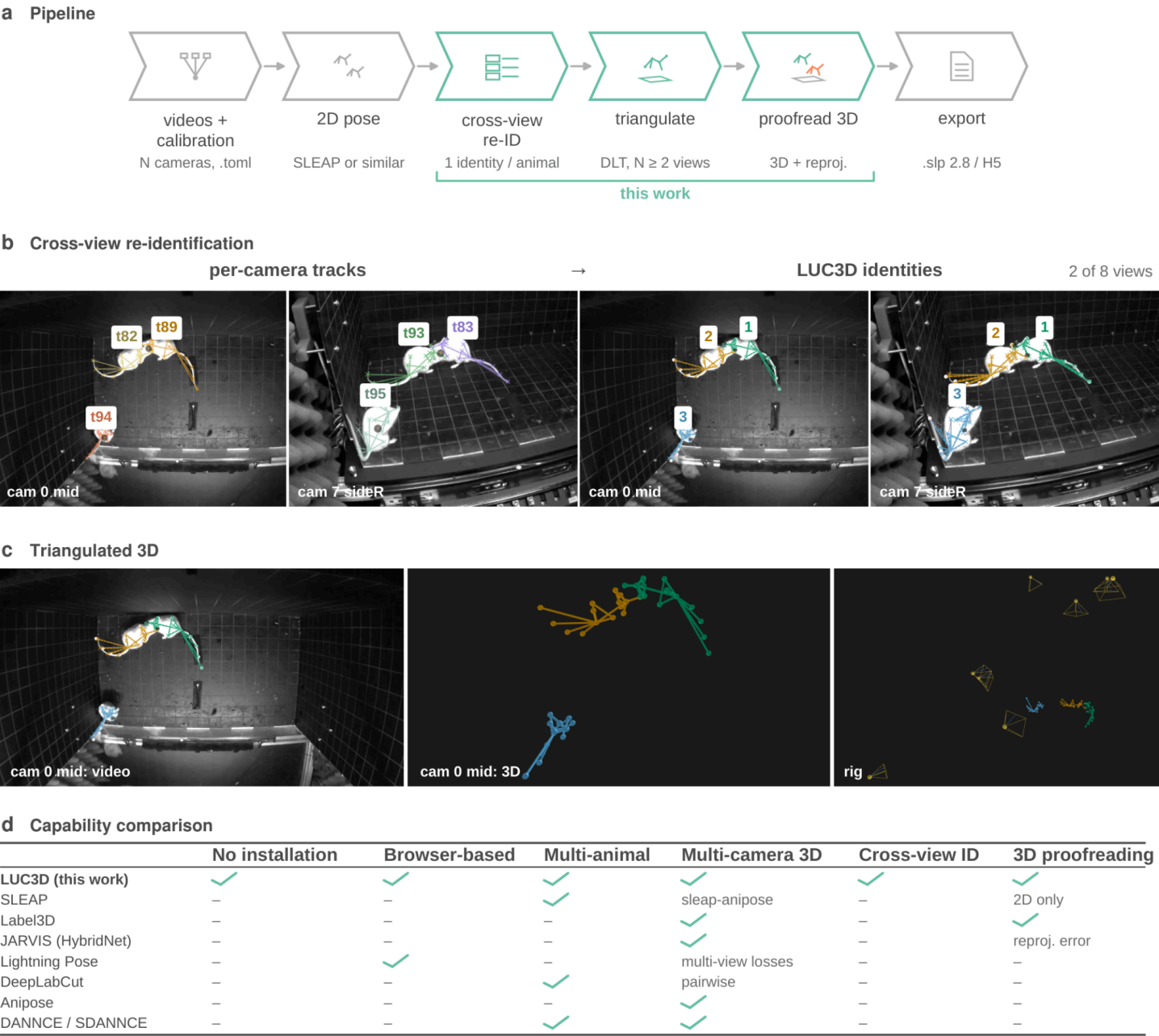
Projects are written as SLEAP .slp files, including the columnar session data introduced in version 2.8 of that format, which carries the calibration, the cross-view identities and the 3D points, so that a project

round-trips through SLEAP and through this application without loss. Labels can also be exported as JSON, 3D points as HDF5, and the 3D viewport as MP4.

Software

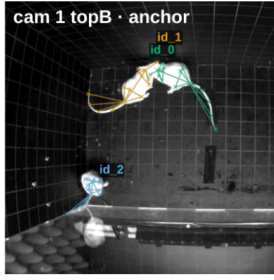
The application is written as vanilla ES modules with no build step. It reads and writes version 2.8 of the SLP format through a vendored copy of `sleap-io.js` and HDF5 through a vendored copy of `h5wasm`, and renders in three dimensions with `Three.js`. A recent Chromium-based browser is recommended, because frame-accurate decoding uses `WebCodecs`. The benchmark drivers run the unmodified application modules under Node 26, so the association and triangulation measured here are the shipped code paths rather than a reimplementations of them. Analysis and figures were produced in Python 3.12 with NumPy, pandas, h5py, SciPy and Matplotlib. Tracking metrics come from `motmetrics` and the comparison triangulator from `aniposelib` 0.7.2.

Fig 1. System overview

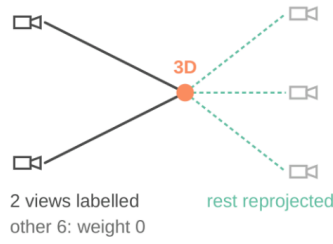


a Protocol

1. Label 2 anchor views



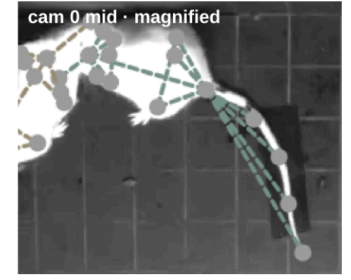
2. Triangulate



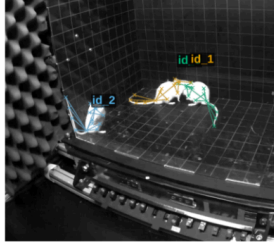
3. Reprojections appear



4. Accept or nudge



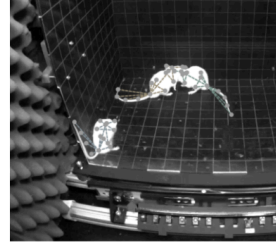
cam 6 sideL · anchor



3D from the 2 anchors



cam 2 topC · not labelled



2-anchor solve, this frame

anchor views 1.4–4.5 px

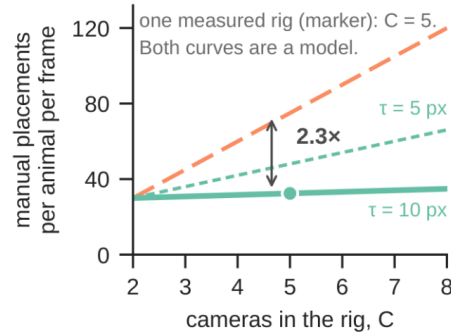
the other 6 2.5–24.6 px

dotted = reprojected

solid = detected

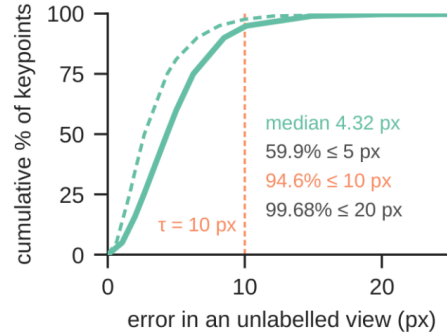
b Placements vs rig size

traditional (assumed)
reprojection-aided



c Cost of two anchors

— vs the view's own detection
-- vs the reference 3D



d Anchor-pair geometry

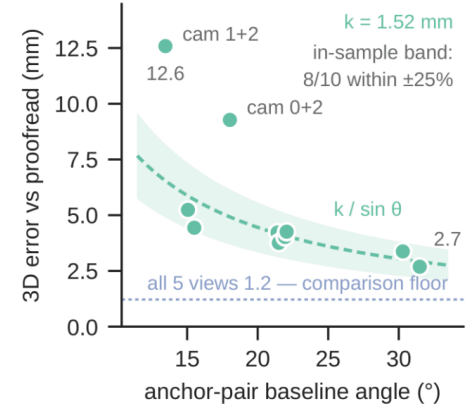
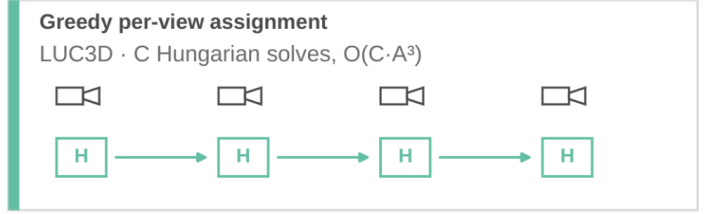
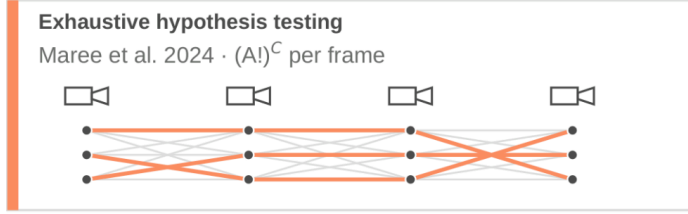


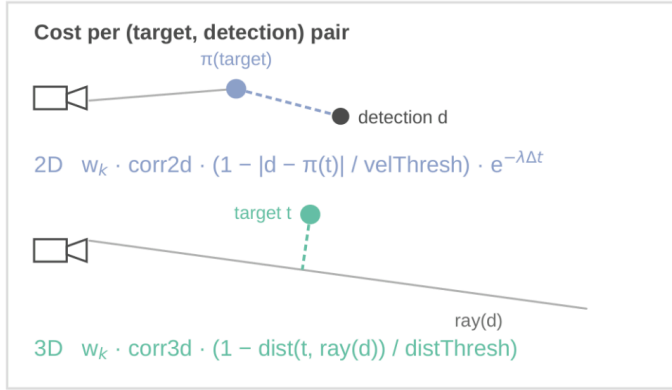
Fig. 2) A) The protocol in the luc3d GUI: two anchor views labelled (cam 1 topB, cam 6 sideL); the 3D solved from those two alone; the resulting reprojection drawn into the remaining views (cam 0 mid and cam 2 topC, neither labelled); and one view magnified beside that frame's measured per-view error split. **B)** Manual keypoint placements per animal per frame against rig size C , for two accept tolerances ($\tau = 10$ px, solid; $\tau = 5$ px, dashed) and for labelling every view. The filled marker is the only measured point ($C = 5$); both curves are the model aided = $2N + (C - 2)Np$ against traditional = CN , with $N = 15$ nodes. **C)** Cumulative distribution of reprojection error in views that were not labelled, scored against that view's own detection (solid) and against the all-view reference 3D (dashed). Median 4.32 px; 59.9% at or below 5 px, 94.6% at or below 10 px. **D)** Median 3D error of a two-anchor solve against the angle the anchor pair subtends at the animal, one marker per camera pair. Dashed curve, the depth-uncertainty law $k/\sin(\theta)$ with $k = 1.52$ mm; band, plus or minus 25%; dotted line, the all-five-view floor (1.2 mm). $n = 50$ sessions, 1,277,424 keypoints, 5 cameras, 2 mice, 15 nodes; every session enters every panel.

Fig 3. Cross-view tracking algorithm

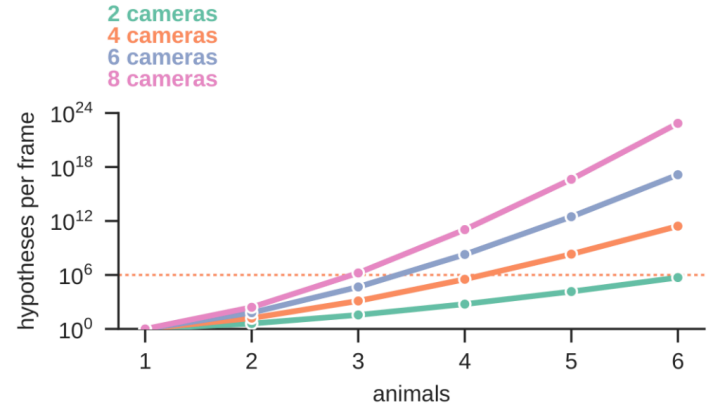
a Grouping strategies



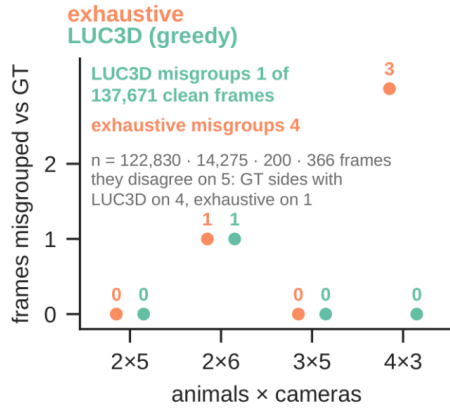
b Association cost



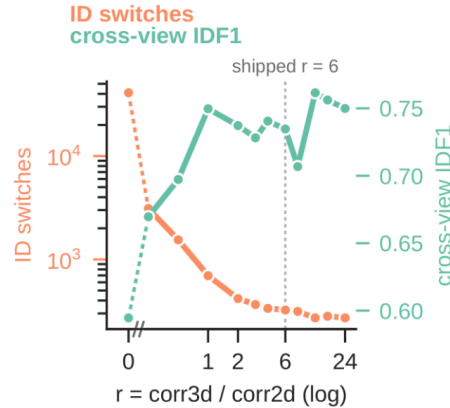
c Hypotheses per frame



d Grouping accuracy, head to head



e 3D-term ablation



f Time per frame

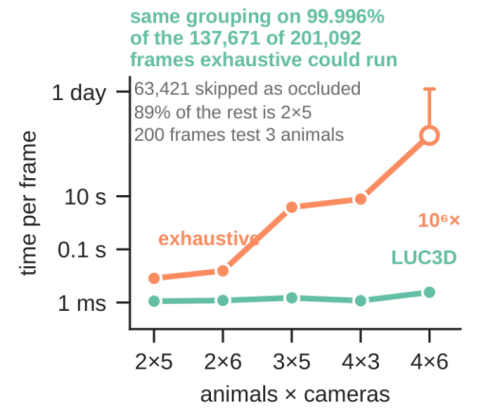


Fig. 3) A) The two search strategies. Exhaustive hypothesis testing (Maree et al., 2024) enumerates every grouping of detections into identities, $(A!)^C$ per frame for A animals and C cameras, and keeps the lowest reprojection error. LUC3D solves one Hungarian assignment per camera, committing each camera before the next, at $O(C \cdot A^3)$. **B)** The association cost for each target and detection pair: a 2D term on the distance between the detection and the target's reprojection into that view, decayed by target age, and a 3D term on the distance from the target to the detection's back-projected ray. Schematic. **C)** Hypotheses per frame for exhaustive enumeration, one curve per rig size, exact arithmetic. Dotted line, the harness cap of 10^6 hypotheses per frame. **D)** Frames whose grouping differs from proofread ground truth, per configuration and method, on identical detections and on every frame exhaustive enumeration could run (137,671 frames in which every camera holds exactly A detections). LUC3D misgroups 1 frame and exhaustive 4. On the 5 frames where the two methods choose different groupings, ground truth agrees with LUC3D on 4, exhaustive on 1. **E)** Ablation of LUC3D against itself: 24 combinations of the two cost weights, which collapse onto their ratio $r = \text{corr3d} / \text{corr2d}$. Cross-view IDF1 (right axis) and summed within-view ID switches (left axis, log scale). $r = 0$, the 3D term switched off, sits left of the axis break. The shipped default, $r = 6$, is marked. **F)** Measured time per frame for both methods on identical detections. The 4-animal, 6-camera configuration exceeds the cap and ran zero frames; its open marker is an arithmetic lower bound and the bar reaches its as-published value.

Fig 4. Triangulation accuracy

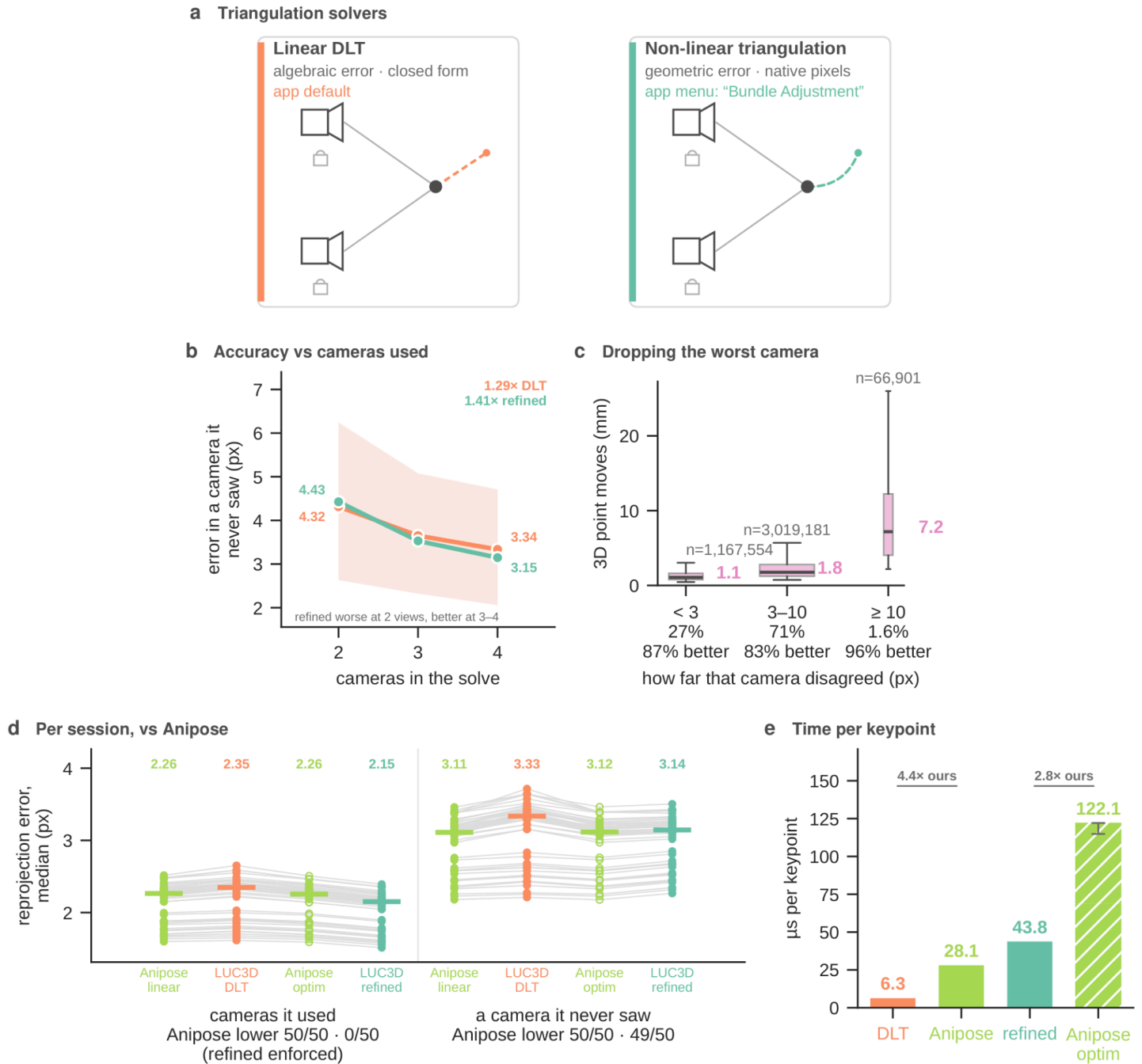


Fig. 4) A) The three solvers. Linear DLT, closed form, the app default; non-linear triangulation, geometric error in native pixels with the cameras held fixed; and joint bundle adjustment, which frees the cameras, present in the code and not exposed in the interface. Padlocks mark fixed cameras, arrows free ones. **B)** Error in a camera outside the solve, against the number of cameras used, for both LUC3D solvers. Every C-choose-k subset; median over 50 sessions; band, the median per-session p25 to p75 for the DLT. DLT 4.32, 3.65 and 3.34 px and refinement 4.43, 3.53 and 3.15 px at 2, 3 and 4 cameras. $n = 31.9, 21.3$ and 5.3 million held-out measurements. **C)** Distance the 3D estimate moves when the worst-fitting view is dropped, by how far that view sat from the all-view solution. Medians 1.07, 1.76 and 7.18 mm ($n = 1,167,554, 3,019,181$ and $66,901$). Boxes, p25 to p75; whiskers, p5 to p95; box width proportional to the square root of n . **D)** Median reprojection error per session for four solvers, paired by algorithm class; one dot per session, rules at the median of the 50 sessions, the same 4,253,636 keypoints in every column. Left, scored in the cameras the solve used; right, in a camera no solve saw, where Anipose is lower in both pairs (50 of 50 and 49 of 50 sessions). **E)** Solve time per

keypoint, with undistortion excluded from all four measurements. LUC3D is 4.4 times faster on the linear pair and 2.8 times on the non-linear pair.

Fig 5. Behavioral analysis of social rearing behavior.

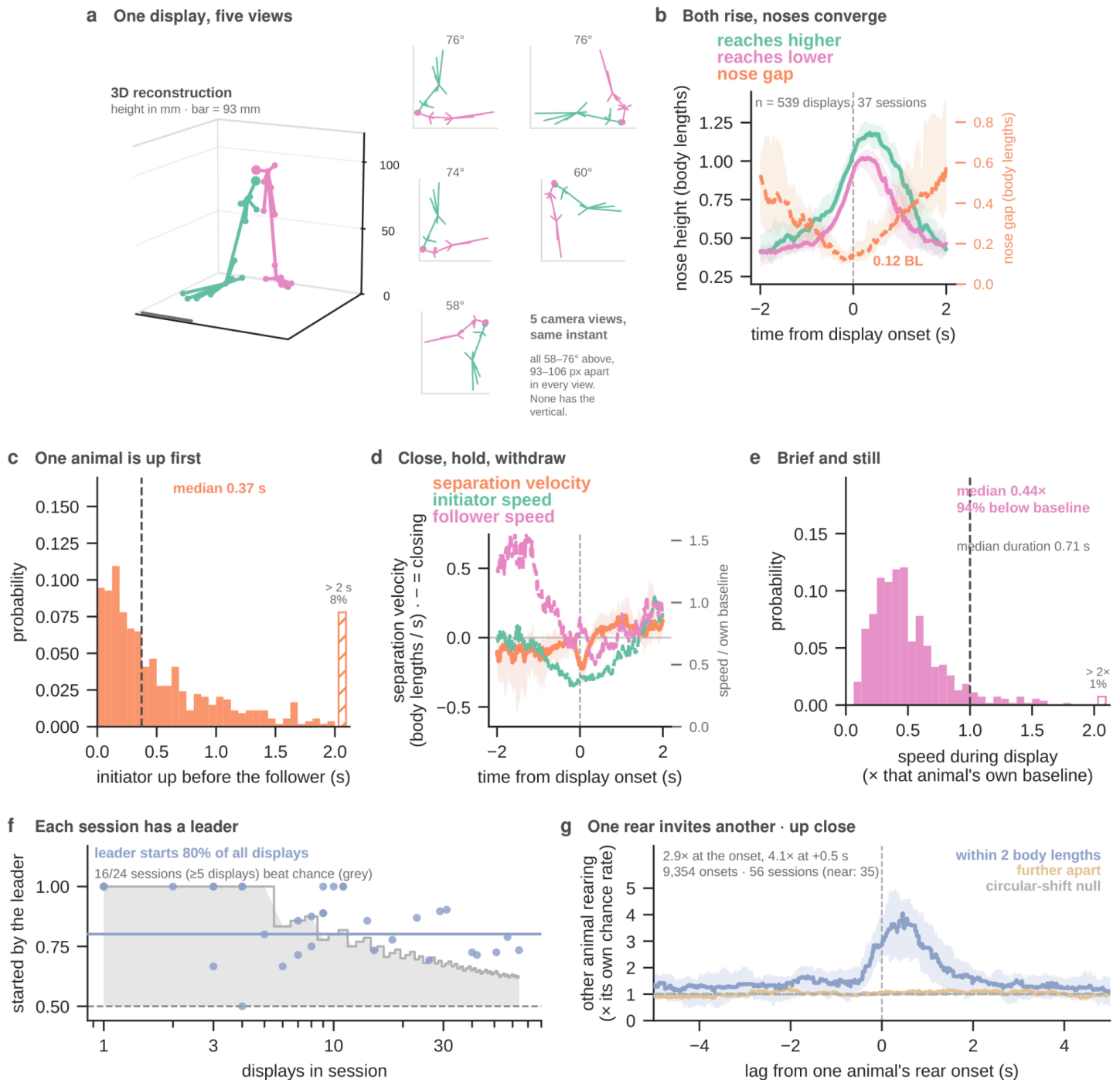
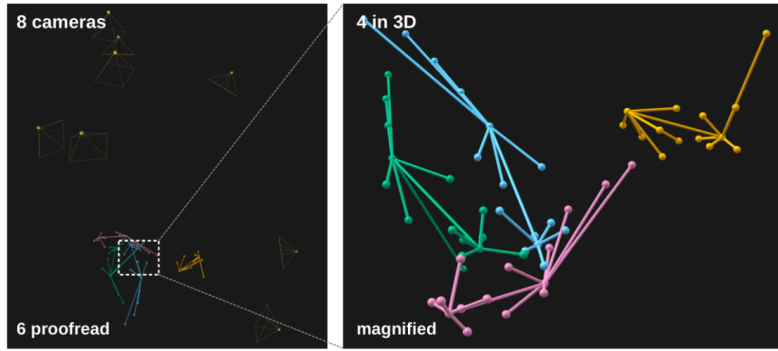


Fig. 5) Two mice rear together face to face, and one animal of each pair starts 80% of the displays. **A)** One display in five views: the 3D reconstruction and the same instant as each camera saw it. Every camera sits 58 to 76 degrees above the animals, so the height that defines the event exists only after triangulation. Scale bar, 93 mm, one body length. **B)** Time course around display onset: across-session median of per-session median curves, band p25 to p75 across sessions. The two height curves are within-display ranks, not individuals. The nose gap falls to 0.12 body lengths at onset. **C)** Time the initiator is up before the follower joins, as a fraction of all 539 displays. Median 0.37 s (p25 to p75, 0.16 to 0.89 s). Hatched bar, overflow: 7.8% of displays exceed 2 s. **D)** Separation velocity of the two tail bases (left axis; negative is

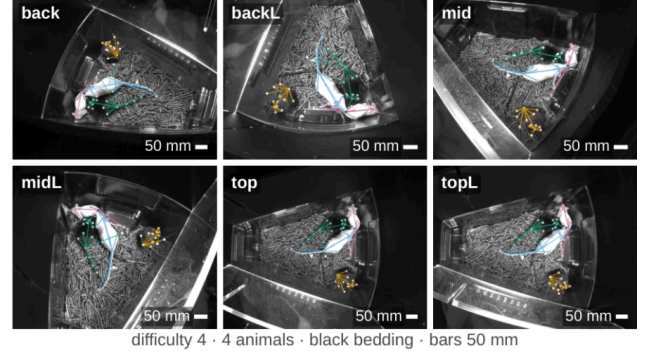
closing) and each animal's speed relative to its own baseline (right axis), by role. **E)** Speed during the display as a multiple of that animal's own session median. Median 0.44, with 94% of displays below baseline and a median duration of 0.71 s. Hatched bar, overflow. **F)** Fraction of each session's displays started by its leader, the animal that started more of them, against the number of displays in the session; one dot per session, $n = 37$. Blue line, the pooled figure over all displays, 432 of 539 (80.1%). Grey band, the central 95% of leader shares a fair coin gives at that session size. 16 of the 24 sessions with 5 or more displays clear the band; among those, median leader share 0.83 (IQR 0.73 to 0.89), Wilcoxon signed-rank against 0.5, $P = 9 \times 10^{-6}$. **G)** For every rear onset by one animal, the probability that the other animal is rearing at each lag, divided by that animal's own base rate; 1.0 is chance. Lines, across-session medians; band, p25 to p75. Within 2 body lengths the curve reaches 2.9 at the onset and peaks at 4.1 half a second later; beyond 2 body lengths it is flat at 1.05; the circular-shift null is flat at 0.99. $n = 539$ displays from 37 of 56 sessions (2 mice, 5 cameras, 150 fps). A display is both animals reared, neck above 0.75 of that animal's own body length, tail bases within 2 body lengths, held for at least 0.25 s.

Fig 6. Multi-animal mouse social behavior datasets

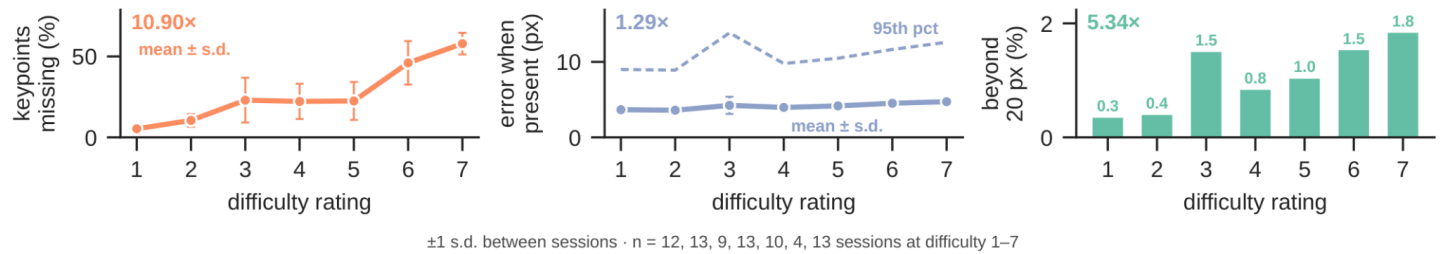
a Rig and 3D



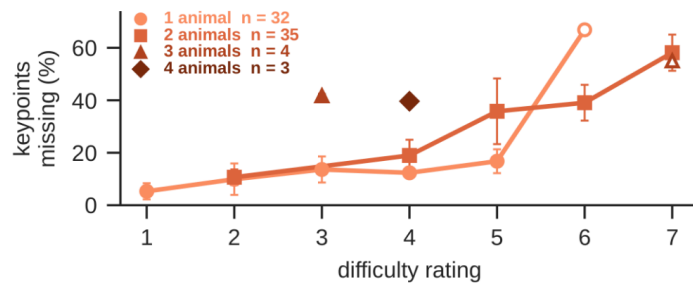
b One frame, six cameras



c Detection quality



d Animal-count control



e Corpora

	BMimica	SLAP-2M	Total
Cameras	5	8 (6 proofread)	—
Animals	2	1–4	1–4
Sessions with 3D	56 of 56	74 of 84	130 of 140
Frames	10,084,734	1,954,440	12,039,174
Hours	18.7 @150 fps	10.9 @50 fps	29.5
Nodes	15	15	15
Measured in c, d, f	no	yes	no

f Difficulty strata

Difficulty	Sessions	Keypoints	Animals	Bedding (black/white)	Missing (%)	Error mean (px)	Error p95 (px)	Error p99 (px)	> 20 px (%)
1	12	252,265	1	12/0	5.3	3.67	9.00	14.52	0.34
2	13	342,287	1, 2	8/5	10.5	3.60	8.91	14.67	0.39
3	9	229,072	1, 3	4/5	23.0	4.24	13.85	21.58	1.50
4	13	245,731	1, 2, 4	6/7	22.2	3.97	9.78	19.04	0.83
5	10	273,748	1, 2	6/4	22.5	4.17	10.49	20.88	1.03
6	4	91,922	1, 2	1/3	46.0	4.52	11.65	24.61	1.53
7	13	126,890	2, 3	7/6	57.9	4.72	12.60	27.36	1.83

Fig. 6 Session difficulty acts on whether a keypoint is detected rather than on how well: the per-view miss rate rises 10.9-fold across strata (5.3% to 57.9%) while the mean error of the detections that do fire rises 1.29-fold (3.67 to 4.72 px). **A**) The rig as LUC3D renders it, all 8 cameras with the frame's own reconstructed animals, with the boxed cluster of 4 animals magnified at the right. **B**) One frame across the 6 proofread cameras, cropped to the app's per-view bounding box, instances coloured by identity. Scale bars, 50 mm. **C**) Raw per-camera detection quality against the difficulty rating, in three sub-plots: keypoints missing (10.90-fold), error of those present (mean plus or minus s.d. between sessions, with the 95th percentile; 1.29-fold), and the fraction beyond the app's 20 px tolerance (5.34-fold). n per stratum is printed under the row. **D**) The same miss rate re-run within each animal count (n = 32, 35, 4 and 3 sessions for 1 to 4 animals). Error bars, plus or minus 1 s.d. between sessions; hollow markers, cells resting on a single session. **E**) Corpus composition, attributes down and corpora across. **F**) Per difficulty stratum: sessions, keypoints, animal counts, bedding split, error mean, p95 and p99, and the fraction beyond 20 px.

Fig 7. Supplemental

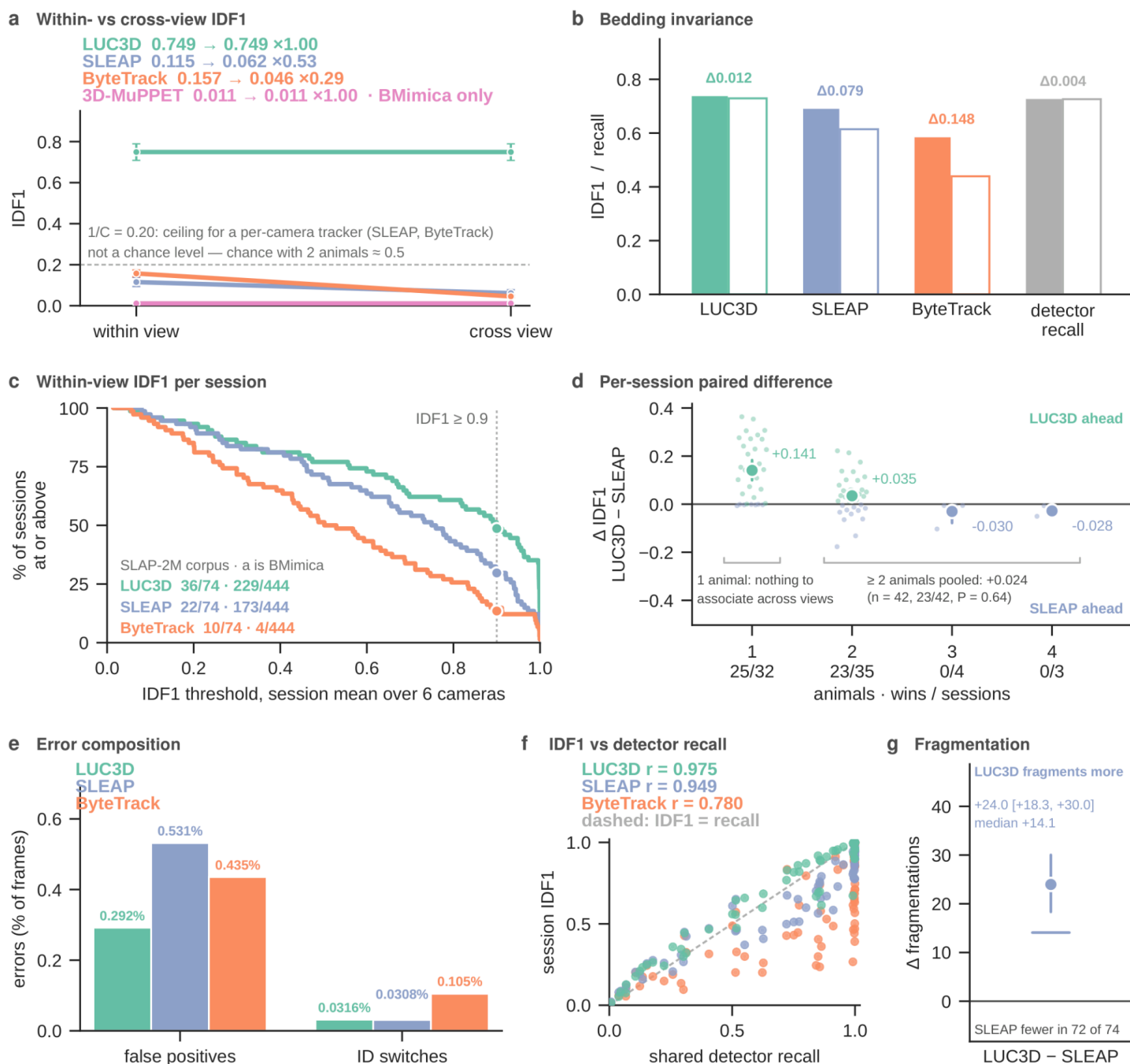


Fig. 7) On identical detections, LUC3D is the only method whose identities survive being pooled across cameras: its IDF1 is unchanged from within-view to cross-view scoring (0.749 to 0.749) while per-camera trackers lose half to three quarters of theirs. **A)** Within-view against cross-view IDF1 for four trackers over 50 sessions; mean with 95% bootstrap CI over sessions at both ends. The line at $1/C = 0.20$ is the ceiling for a tracker that labels each camera independently, not a chance level; with 2 animals chance is near 0.5. **B)** IDF1 under black bedding (solid) and white bedding (open), with the shared detector's own recall as a control. This is a between-session comparison, 44 black against 30 different white sessions, so the bars are not paired and carry no intervals. **C)** Within-view IDF1 over 74 sessions as a survival curve, one step per session. At IDF1 of 0.9 or above the counts are 36, 22 and 10 of 74. **D)** Paired per-session difference in within-view IDF1, LUC3D minus SLEAP, by animal count; mean with 95% bootstrap CI, every session drawn behind its cell, n and win count under each tick. Pooled over all 74 sessions the difference is +0.075, carried by the 32 single-animal sessions (+0.141, 25 of 32, sign test $P = 0.002$); over 2 or more animals it is +0.024 (n = 42, 23 of 42, $P = 0.64$). **E)** False positives and ID switches as percentages of camera-frames (11,726,640 camera-frames). False negatives are 98.8 to 99.3% of every tracker's error budget and are not plotted. **F)** Session within-view IDF1 against the shared detector's recall,

one point per session, with the diagonal on which the two are equal. LUC3D $r = 0.975$, SLEAP $r = 0.949$, ByteTrack $r = 0.780$. **G)** Paired fragmentations per camera-session, LUC3D minus SLEAP; mean with 95% bootstrap CI over 74 sessions, with the median beside it. +24.0 (95% CI +18.3 to +30.0), median +14.1; SLEAP fragments fewer in 72 of 74 sessions. A uses 50 BMimica sessions (5 cameras, 2 mice); B to G use 74 SLAP-2M sessions (6 cameras, 1 to 4 animals).

Supplemental Screenshot

